CONTINUATION — Helix Proxy: VPN-Aware Dynamic Routing Extension

Revision: 25 **Last modified:** 2026-07-02T16:33:00Z **Status:** Active — **Rev-25 wave (2026-07-02): two operator directives DOCUMENTED. (Part 1) Mullvad WireGuard egress — COMPLETE:** a persistent ONE-DEVICE Mullvad WireGuard identity was registered against the operator's Mullvad account via the current app API (`POST /auth/v1/token` → `POST /accounts/v1/devices`; the legacy `/wg` endpoint is deprecated), device "groovy rabbit" (id `df4184d4-b6e9-4c2e-b7f7-16e15d4e55a2`), creds stored in the gitignored `.env` (`VPN_SERVICE_PROVIDER=mullvad` + WireGuard key + assigned address — §11.4.10; §11.4.10.A pre-store leak audit CLEAN, 0 tree/history leaks). **LIVE EGRESS PROVEN:** gluetun in rootless podman brought up a kernelspace WireGuard tunnel (relay `cz-prg-wg-101`) and `am.i.mullvad.net/json` returned `mullvad_exit_ip=true` (exit IP `146.70.129.117`, Prague, Czech Republic); "one same device" is guaranteed by reusing the persistent private key; host routing untouched (§11.4.174), verification container torn down (§11.4.14). Evidence: `qa-results/verification/mullvad_egress_20260702T161312Z/PROOF.txt`. This RESOLVES the P10 real-VPN-egress §11.4.21 operator gate (`docs/design/hardening/Status.md` Rev 7); the full proxy-data-plane-through-gluetun e2e + up→drop tcpdump killswitch half remain the next step (§3 OWED-TO-P10). **(Part 4) live DNS DEFERRED by the operator (2026-07-02):** Let's Encrypt Phase 4/6 (live HTTPS issuance/renewal/rotation) stays **OPERATOR-BLOCKED, reason "operator-deferred DNS (2026-07-02)"**, unblock = operator-provided LE account + DNS credentials (`docs/design/letsencrypt/Status.md` Rev 4); the hermetic Phase-3/5 issuance/rotation guards remain GREEN — only the live-cert half is deferred. **(prior) §11.4.126 autonomous loop** (operator: "keep hardening, don't tag yet" + "3-4 parallel subagents, rock-solid evidence, no bluff" + **"all future work on `main`"**). **Rev-24 wave (HEAD `860d38e`, branch `main`, FF to origin): anti-bluff audit ledger CLOSED + core evidence library provably audited.** `c44e90c` **#74/#75** (F-C `vpn_failclosed` all-timeout SKIP reason → honest inconclusive; F-D `memory_soak` degenerate-final-RSS-sample → honest SKIP not "bounded" — each §11.4.135-guarded, single-source, conductor-inline-reviewed). `860d38e` **#76 (F-E, MEDIUM)** — an INLINE §11.4.118 audit of the core `tests/lib/evidence.sh` (which the dynamic-audit had EXCLUDED) found +

DEMONSTRATED a THIRD absence-as-evidence bluff at the LIBRARY ROOT: `assert_no_leak` scored an EMPTY/malformed capture as "no leak" PASS (reachable via `no_leak_analyzer.sh:62`); fixed FAIL-closed (empty→FAIL; fallback branch requires positive tcpdump structure) with NO false-FAIL (a genuine tcpdump-footer-0 still PASSes) + 3 selftest negatives (48/48) + a §11.4.135 guard `assert_no_leak_empty_capture_test.sh`, conductor-inline-reviewed. **Then audited the WHOLE core evidence library** (§11.4.118 completeness): 8 verdict helpers — `ab_pass_with_evidence` (empty/absent→FAIL), `ab_skip_with_reason` (invalid reason→FAIL rc2), `proxy_conn_verdict` (timeout can't PASS), `wg_transfer_delta` (flat/missing→FAIL), `assert_egress_ip` (unverifiable half→OPERATOR-BLOCKED not PASS, §11.4.68/F7), `assert_cache_hit` (requires access.log `TCP_*HIT`), `assert_graceful_503` (timeout/blank/PID-change→FAIL) all verified fail-closed-SOLID; only `assert_no_leak` had a bluff (F-E, now fixed). **AUDIT LEDGER COMPLETE — SIX findings:** F1 (#71/#72) · F-A/F-B (#73) · F-C/F-D (#74/#75) · F-E (#76), all §11.4.118-discovered → RED-first fixed → §11.4.135-guarded (a standing regression suite of 4 new guards: security-group-critical-gate, chaos-no-leak-argshape, vpn_failclosed-reason, memory_soak-degenerate, assert_no_leak-empty). ⚠ **Throttle oscillates:** saturates even SOLO dispatches when the rolling rate window is full, recovers after ~10min of conductor-inline (no-dispatch) work; **conductor-inline review is throttle-immune AND a valid §11.4.70/§11.4.142 reviewer for subagent-authored fixes** (conductor ≠ builder subagent) — it reviewed #73/#74/#75/#76 + captured #69 this session. **The non-operator-gated actionable queue is now EXHAUSTED** (all ledger tasks done, audit ledger closed, evidence library provably clean). Remaining is ALL operator-gated: live svord bridge (biggest unlock — flips ~16 protocol SKIPs to live PASS + a full data-plane e2e), LE #59, podman aardvark-dns compose-network, real-device flash, HelixQA 6-sibling vendoring. No tag ("keep hardening"). **Rev-23 wave (HEAD `3ec39d3`+doc, branch `main`, FF to origin): #73 (second demonstrated bluff) CLOSED + #69 RESOLVED.** (a) Solo dynamic-audit `ac3f1c89` found **F-A**: `chaos_suite.sh` C1 no-leak proof was VACUOUS — the call site passed a full URL to `no_leak_analyzer` which greps tcpdump `IP .*<t>` (IPs, never scheme-prefixed URLs) → leak count always 0 → no-leak PASS regardless of a REAL leak (same §11.4.107(10) class as F1); + **F-B** empty-capture silently skipped the analyzer. Fixed RED-first `3ec39d3` (`chaos_target_leak_key` resolves URL→host→IP; `chaos_leak_signal` rc 2 UNEVALUATED→honest SKIP; §11.4.135 guard `chaos_no_leak_argshape_test.sh`, RED_MODE polarity, single-source via `CHAOS_SOURCE_ONLY`). Reviewed INLINE by the conductor (the review subagent crashed on the throttle; conductor ≠ builder-subagent author → valid §11.4.70/§11.4.142); guard load-bearing under sh+bash; no_leak_analyzer.sh unchanged. F-C/F-D tasked LOW (#74/#75). (b) **#69 RESOLVED (§11.4.6 correction):** ran `go test -run Integration -v ./...` on control-plane → **14 real PASS / 1 honest §11.4.3 SKIP / 0 FAIL** against REAL podman

Postgres/redis/gluetun (evidence `qa-results/integration/control-plane_verbose_20260702T125403Z/`). A prior round had wrongly marked Integration PENDING assuming podman was unusable — FACT: the aardvark-dns break is compose-network-only, NOT ad-hoc `podman run`. Hardening matrix (Status Rev 6): **Integration PENDING→PASS**; **E2E PENDING→honest §11.4.3 SKIP** (no in-project e2e suite; end-to-end covered by the P10 fail-closed proof + the integration path; a full compose-network-wired stack e2e is operator-gated). ⚠ **THROTTLE now saturates even SOLO dispatches** (~16 subagents this session → rolling rate window full); shifted to conductor-INLINE work (throttle-immune) — which independently reviewed #73 AND captured #69. **Next:** #74/#75 (LOW) when the rate cools. No tag ("keep hardening"). **Rev-22 wave (HEAD `159fcbc`, branch `main`, FF to origin): #67 COMPLETE — Go control-plane §11.4.169 resilience gap closed** in 3 independently-reviewed slices: DDoS `0c51f61` (loadflood + decide-flood; calibrated-per-run p99 + goroutine/fd leak census + Type-4 torn-read fold), memory-soak `ceb4839` (growth-ratio N-invariant 1.11-1.13; caught+fixed a fake-audit-accumulation false-fail via `drainFakeAudit`), chaos `159fcbc` (store-drop-mid-`WithTx` rollback / SSE-disconnect reap / ctx-cancel honest-SKIP / redis-unavailable → zero `(true,*)` leaks across the outage). All in-process httptest — no Postgres/redis/podman/netns; each §1.1-load-bearing (reviews a02f558f / a74e23fd / a139f9ec GO). Concurrency was already covered (concurrency_test.go + atomicity_test.go + `-race`) — the #67-design correctly avoided a duplicate suite (§11.4.6/§11.4.27). **§11.4.6 honesty update:** the hardening §11.4.169 matrix's green DDoS/chaos/memory rows had cited DATA-PLANE Squid evidence ONLY — the Go server had no such coverage; matrix now updated (`docs/design/hardening/Status.md` Rev 5 + Status_Summary Rev 5) to reflect the new Go-control-plane coverage, and the F1 fix+guard recorded in the hardening Security row. ⚠ **Throttle: STRICTLY single-stream, confirmed 3× incl. the minimal 1+1 case** — any subagent dispatched while another runs crashes on the server-rate throttle; done retesting (retests just waste dispatches). Effective parallelism = 1 subagent + conductor-inline work. **Next (single-stream):** #69 (control-plane integration/e2e evidence re-capture, §11.4.40 pre-tag — likely operator/podman-gated → honest §11.4.3 SKIP) + respawn the dynamic-suite bluff-audit (§11.4.118, throttle-crashed 3×). No tag ("keep hardening"). **Rev-21 wave (HEAD `5771a46`, branch `main`, FF to origin):** four more single-stream landings + an anti-bluff win. `c32dcad` healthd coverage 70.3→79.3% (run() fail-closed + `-version` tests, review `a07d4ca4` GO, load-bearing). **F1 arc (§11.4.118→§11.4.6→§11.4.135):** a tight-scoped bluff-audit (`a1bf492d`) found `proxy_acl_security.sh` aggregated `≥1 PASS && 0 FAIL ⇒ group PASS` — so a non-critical S3-only pass reported the group GREEN *naming S4 "GREEN live"* while the security-critical S1/S4 SKIPped (plausibly LIVE on the broken-podman host); `6e3e031` **FIX** (group PASS now requires S1&&S4 to actually pass, else SKIP naming the absent one; run-tests headline honest;

review `aefcdb5d` GO, adversarial-revert-proven) → `adca02e` **§11.4.135 standing guard** (single-source `tests/lib/acl_group_verdict.sh` + RED_MODE regression test wired into the suite; review `a5ff11ac` GO — reverting the lib gate flips the guard to FAIL). `5771a46` **#68** wired the 2 orphaned harnesses (`container_boot.sh` + `discovery_reflect.sh`) into the Challenge bank (both honest-SKIP, runner RESULT OK total=8/SKIP=8, review `ab1783f4` GO) — §11.4.135 orphan gap closed. ⚠ **Throttle STILL caps concurrency (§11.4.6, empirically RE-tested):** dispatching 2 new subagents concurrent with 1 running → BOTH new crashed on the transient throttle at ~25s while the pre-existing single stream survived + GO'd. So **single-stream sequential** remains the sustainable pace; 3-4 parallel is environment-blocked right now, not a choice. **Pending (single-stream):** respawn #67-design (control-plane resilience blueprint) + the dynamic-suite bluff-audit (both throttle-crashed); then #67 build; #69 (integration evidence, likely operator/podman-gated). No tag ("keep hardening"). **Rev-20 wave (HEAD `85d8b32`, branch `main`, FF to origin): #65 sniff fan-out LANDED.** `85d8b32` replicates the underlay-sniff AF_PACKET non-leak differential onto 3 protocol harnesses (`hermetic_bridge_run.sh` marker `$DEV_NAME` / `_ftp_run.sh` `$NONCE` / `_webdav_run.sh` `$NONCE`) — a verbatim single-source clone of the ethertype-guarded substrate `_emit_an_py` analyzer; NORMAL `ciphertext(0x04:51820)=present`+`marker-absent`, `SNIFF_MUT=plain` load-bearing (only plaintext-absent flips, ciphertext stays present); the `:9` discard port stays distinct from each §11.4.111 TCP negative-control port. **Email is honest N/A** (implicit-TLS encrypts the token *below* WG → a plaintext-absent underlay sniff is tautologically true regardless of the tunnel, a §11.4-forbidden vacuous test; N/A note in the harness header + `docs/scripts/hermetic_email_run.md`). Provenance (§11.4.147): the builder `a9cb5106` crashed on a transient server throttle at ~13 min, but its work was **conductor-verified inline** (preserve→resume — all 3 NORMAL PASS + `SNIFF_MUT=plain` load-bearing + original teeth regression-clean) then **independently reviewed `a1dca6fd` GO 7/7**. This doc-sync marks FINDINGS §7.1 IMPLEMENTED + adds §11.4.18 sniff notes to the 3 harness companion docs + Status pair → Rev 9. ⚠ **Throttle note (§11.4.6):** a server-side transient throttle ("Server is temporarily limiting requests — NOT your usage limit") crashed **3 of 4** concurrent subagents (#65 builder, bluff-audit, #67-design) → dropped to **single-stream sequential** dispatch (a single subagent survives; a 4-wide fan-out throttles). **Pending (single-stream as the throttle allows):** Go-coverage-healthd review (`a0f6ad3` GO'd — `control-plane/cmd/healthd/main_lifecycle_test.go` raises healthd 70.3→79.3%, untracked, awaiting review→commit); respawn the crashed shell-bluff-audit + #67-design; then #67/#68/#69. No tag ("keep hardening"). **Rev-19 wave (HEAD `172eb5c`, branch `main`, FF to origin): session-limit reset → landed #66 + #70 (both independent §11.4.142 GO).** `b66b2fa` **#66** wired `hermetic_email_run.sh` into the `test_vpn_lan_hermetic` standing guard — closing the §11.4.6/§11.4.135 doc-bluff the §11.4.169 ledger audit caught (email was `refs=0` in run-tests.sh while the docs claimed all four wired); independent review `a5bc6266` GO 6/6, the REAL loop (temp copy inside tests/ for correct readonly SCRIPT_DIR) shows ALL 5 rows PASS incl `VPN-LAN hermetic: Email … PASS`, verdict-map adversarially confirmed to send a harness FAIL→loop-FAIL (never silent-SKIP), honest-SKIP intact; docs re-corrected to "4 wired". `172eb5c` **#70** adopted the found orphan test `control-plane/cmd/acl-helper/main_run_test.go` (§11.4.124 — a prior session's uncommitted coverage PWU, never debris; covers `run()` fail-closed Redis-connect + `main() -version`); independent review `aae14631` GO — race-clean, 80.8% pkg coverage, LOAD-BEARING (both fail-open mutations of `run()`'s guard kill the test). Session-limit reset was awaited via a backgrounded `until` clock-wait (autonomous, no operator round-trip). **In flight (parallel):** #65 sniff fan-out builder (`a9cb5106`, netns owner — 3 harnesses bridge/ftp/webdav + honest email N/A note per FINDINGS §7.1) + shell-bluff-audit respawn (`a67c0093`, read-only §11.4.118). **Next:** review+land #65 (→ FINDINGS §7.1 implemented + Status sniff rows); triage bluff-audit findings; then Go-coverage respawn + #67/#68/#69. No tag ("keep hardening"). **Rev-18 wave (HEAD `cdb0ccd`, branch `main`, FF to origin): #64 ethertype guard LANDED + 4-parallel-stream fan-out + a §11.4.169 ledger audit that caught a real doc-bluff (corrected).** `cdb0ccd`: a one-line Ethernet-ethertype guard on the underlay-sniff frame analyzer (skip non-`0x0800` frames — VLAN offset shift, structurally impossible on the fresh veth, correct-by-construction), refactored to single-source `_emit_an_py()`/`scan_frames()` so a `--selftest-analyzer` exercises the SAME parser; independent §11.4.142 review `af9f67ad` = GO 5/5 with the decisive load-bearing proof (WITH guard `vlan_ct=absent` PASS; WITHOUT, scratch-neutralized, `vlan_ct=present` FAIL). Per the operator directive, dispatched **4 parallel streams** (§11.4.103): #64 review (GO→landed); a #65 fan-out **DESIGN** (§11.4.150) that PREVENTED a bluff — the sniff is meaningful for **3** harnesses (bridge/ftp/webdav, plaintext-under-WG) but **N/A for email** (implicit-TLS encrypts the token *below* WG, so "plaintext absent on the underlay" is tautologically true regardless of the tunnel — a §11.4-forbidden vacuous test); a §11.4.169 test-type **ledger reconciliation** (COMPLETED, real findings); and Go-coverage + shell-bluff-audit streams (both **CRASHED on session limits** — §11.4.147 incomplete-not-done, respawn after the ~09:30Z reset, no work lost). **The ledger surfaced a genuine documentation bluff (§11.4.6/§11.4.135/§11.4.138):** `docs/features/vpn_lan/Status.md` claimed all FOUR hermetic protocol legs are wired standing-suite guards, but `hermetic_email_run.sh` is `refs=0` in `run-tests.sh` (conductor-verified — only substrate+Cast/FTP/WebDAV wired at :1006-1009). **Corrected the doc to reality this wave** (3 wired + email reviewed-GO-but-wiring-pending) and tasked the actual wiring **#66**; also tasked **#67** (control-plane Go stress/chaos/DDoS/memory §11.4.169 gap), **#68** (orphaned `container_boot.sh` + `discovery_reflect.sh`), **#69** (re-capture integration/e2e evidence, §11.4.40 pre-tag). `FINDINGS.md`→Rev 5 (§7 #64-landed + §7.1 fan-out design), VPN-LAN Status pair→Rev 7, README rows. **Next (session-limit-gated ~09:30Z):** land #65 (3-harness sniff fan-out + honest email N/A note) + #66 (email suite-wiring) with independent review; respawn the crashed coverage + bluff-audit streams (§11.4.147). No tag ("keep hardening"). **Rev-17 wave (HEAD `91af9c6`, branch `main`, FF to origin): §11.4.107 underlay-sniff AF_PACKET non-leak differential landed on the WG substrate.** `91af9c6` adds to `hermetic_wg_roundtrip.sh` a rootless `AF_PACKET` capture on the underlay `veth0` during the positive round-trip asserting BOTH (a) WG **ciphertext present** (type-4 `0x04` datagram to `:51820`) AND (b) the per-run **plaintext nonce ABSENT** in raw underlay bytes — a different-domain oracle (§11.4.107(2)) + the canonical WireGuard self-audit method; the third independent layer of the tunnel-integrity claim after §11.4.111 destination-binding + wrong-destination negative. **Independent §11.4.142 review `a2b3c696` = GO on 11/11 checks against real runs (§11.4.134):** load-bearing golden-bad `SNIFF_MUT=plain` (emits the nonce as cleartext UDP to discard `:9`, distinct from the §11.4.111 TCP `:8080` control) flips ONLY assertion (b) to FAIL 3/3 while ciphertext stays present (not a tautology, §11.4.107(10)); NORMAL → `ciphertext(0x04 :51820)=present plaintext_nonce=absent` 3/3; header-only pcap → analyzer exit 3 (honest FAIL); forced-iface + no-tcpdump → honest `SNIFF-SKIP`; `WG_MUT=badkey` tooth undisturbed (sniff sits past the `UP!=1` gate); veth0-only capture, 3.5 s / 4 MB / `timeout`-bounded, `SNIFF_PID` reaped, `wg0-mullvad` untouched (§11.4.174); `sh -n` + `bash -n` clean. `FINDINGS.md` → Rev 4 §7 (PLANNED → IMPLEMENTED). Two tracked follow-ups: **task #64** one-line Ethernet-ethertype guard on the frame analyzer (reviewer nit — VLAN offset shift, structurally impossible on the harness's own fresh veth, correct-by-construction); **task #65** fan the differential out to the 4 protocol harnesses (bridge/ftp/webdav/email), each with a per-protocol plaintext marker + its own `SNIFF_MUT=plain`. **Anti-bluff §11.4.118 pass this wave:** a mechanical smell-scan of all 26 test scripts (bare `ab_pass`, `|| true`-before-verdict, fail-open SKIP→PASS, unconditional PASS) came back clean — enumerated coverage evidence, with the honest boundary that grep cannot catch a *semantic* tautology (only adversarial re-run can, as in the #62 negative-control fix). **Next: task #64 (tiny) or #65 (fan-out) — the loop continues (no tag — "keep hardening"). Rev-16 wave (HEAD `4dd5fc6`, branch `main`, all FF to origin): §11.4.111 wrong-destination negative control added to all 5 hermetic harnesses + FINDINGS §7 (next hardening) designed.** `4dd5fc6`: after each positive round-trip, the harness probes the peer's service on the UNDERLAY IP `10.9.0.2` and asserts it FAILS (the peer binds the WG-only overlay `10.10.0.2`, so a positive success could only have traversed `wg0` — reachability-as-proof made self-evidencing, not structural-only); a wrong-destination SUCCESS ⇒ fail-closed. **Independent §11.4.142 review ran TWO rounds (iterate-to-GO §11.4.134):** round 1 caught a REAL tautology (§11.4.107(10)) — `hermetic_wg_roundtrip.sh` probed

`/payload.txt` but `rm`'d `$SRVDIR` BEFORE the probe, so a reachable underlay 404'd and `NEG-OK` printed unconditionally (proven by the reviewer binding the peer to `0.0.0.0`: pre-fix the harness wrongly PASSed); fix = defer the `SRVDIR` cleanup to AFTER the control, so post-fix bind-`0.0.0.0` ⇒ `WG_FAIL … unexpectedly served the payload` exit 1 (control now load-bearing), NORMAL still PASS, `WG_MUT=badkey` tooth still fires, 3/3 deterministic; round 2 re-verified ⇒ GO. The other 4 controls (bridge/ftp/webdav/email) were proven load-bearing in round 1 (each FAILs when its peer binds `0.0.0.0`). Part (b) (`/usr/sbin/wg` preflight) was already satisfied in all 5 — verified, no redundant edit (§11.4.6). `FINDINGS.md` Rev 3 §7 designs the next hardening — the **underlay-sniff AF_PACKET non-leak differential** (assert WG ciphertext `0x04` present + plaintext nonce absent on the underlay), cited §11.4.99/§11.4.150, **queued as task #63** (blocked-by #62, now unblocked). Anti-bluff win: a decorative `NEG-OK` that gated nothing was REFUSED by review, not shipped. **Next: task #63 (underlay-sniff) — the loop continues (no tag — "keep hardening"). Rev-15 wave (HEAD `3b73f02`, branch `main`, all FF to origin): H2.email promoted + all hermetic promotions wired into the standing suite as regression guards.** `71e0bac` added `test_vpn_lan_hermetic()` to `tests/run-tests.sh` — the 4 hermetic harnesses (`hermetic_wg_roundtrip.sh`, `hermetic_bridge_run.sh`/Cast-eureka, `hermetic_ftp_run.sh`, `hermetic_webdav_run.sh`) now run on every suite invocation as **§11.4.135 standing regression guards** under a SKIP-aware verdict map (`rc0+PASS≥1+FAIL0⇒PASS; rc0+SKIP≥1+PASS0+FAIL0⇒SKIP; else FAIL`), independent §11.4.142 review GO (7/7 adversarial HOLD; 4/4 real + 7/7 synthetic mapping). `3b73f02` landed **H2.email**: `hermetic_email_run.sh` runs the **UNMODIFIED** `email_roundtrip.sh` autonomously over the tunnel via a pure-stdlib implicit-TLS mail peer (SMTPS 465 / IMAPS 993 / POP3S 995, shared in-memory mailbox) bound to the WG-only overlay `10.10.0.2` (reachability = tunnel-traversal proof §11.4.111). **Independent §11.4.142 adversarial review = GO, no blocking findings (§11.4.134):** NORMAL PASS all 4 scored legs (imaps_login_list, smtp_submission_send, pop3s_retrieve_roundtrip, open_relay_refused; email_reverse_leg SKIP N/A), `MAIL_MUT=openrelay`→real `FAIL: open_relay_refused` (only that leg), `MAIL_MUT=droptoken`→real `FAIL: pop3s_retrieve_roundtrip` (only that leg) — **both teeth load-bearing + targeted**; 3/3 deterministic (§11.4.50); §11.4.10 no cred/key leak (AUTH base64 = server RFC 4954 challenges only); clean cleanup (§11.4.14); `wg0-mullvad` untouched (§11.4.174). The close_notify fix (`tls.unwrap()` before close) + IMAP bare-QUIT clean-close eliminate the silent-SKIP-masking class (§11.4.6) — all legs ran live; builder self-debug (§11.4.102) also fixed the PoC `shutdown()` self-recursion + an IMAP `-ign_eof` hang. `FINDINGS.md`→Rev 2 (**§6 implicit-TLS mail deep-research** §11.4.150: PEP 594 `smtpd` removal, RFC 8314/5321/4954/3501/1939, the close_notify FACT + sources) + companion `hermetic_email_run.md` Rev 1, all HTML+PDF synced 0-fence-leak (§11.4.168). **Honest gate verdict (§11.4.6): the clean zero-install protocol-promotion**

set is now COMPLETE (Cast-eureka + FTP + WebDAV + email). Remaining legs genuinely operator-gated (§11.4.122/§11.4.3): SFTP (no `sshd`), SMB/NFS (no samba/nfsd), `discovery_reflect.sh` scored leg (no `avahi-browse` client), ADB (device), container (podman). **Follow-up (separate reviewed batch): explicit wrong-destination negative (underlay/wg-down MUST-fail) to make §11.4.111 self-evidencing + wg-preflight `/usr/sbin/wg` across sibling harnesses. Next: the loop continues (no tag — "keep hardening"). Rev-14 wave (HEAD `3b98d02`, branch `main`, all FF to origin): H2.x hermetic protocol promotions — three operator-gated protocol legs now run AUTONOMOUSLY over the hermetic kernel-WireGuard tunnel (§11.4.52).** `18a21bd` landed H2 = the UNMODIFIED `chromecast_dial.sh` eureka control leg promoted (`hermetic_bridge_run.sh`, stdlib eureka peer `10.10.0.2:8008`, `H2_MUT=badeureka` teeth, self-fetch name-nonce). `3b98d02` landed **H2.ftp + H2.webdav**: `hermetic_ftp_run.sh` (embedded ~85-line stdlib FTP server `10.10.0.2:2121`, PASV/EPSV both traverse `wg0` §11.4.111, harness content-verifies a self-RETR against ground truth §11.4.107(9), `FT_MUT=empty` teeth) + `hermetic_webdav_run.sh` (stdlib 207 origin `10.10.0.2:8080` reached THROUGH a stdlib forward proxy `127.0.0.1:3128` RFC 7230 §5.3.2→§5.3.1, `WEBDAV_MUT=bad207` = the exact PASS gate). Both harnesses additionally **scrub ambient-shell sibling-leg env vars** before invoking the promoted test (§11.4.50 determinism, independent-review nit — false-negative guard, never a bluff PASS). **All verified GREEN: FTP normal+teeth, WebDAV normal+teeth (207 through proxy, self-fetch body 779 B), both 3/3 deterministic; independent §11.4.142 review returned GO on both.** WebDAV harness was drafted by a session-limit-crashed builder subagent, resumed residue-clean per §11.4.147. Feature Status bumped to **Rev 3 (new §J)** + Status_Summary Rev 3 + companion docs `hermetic_ftp_run.md` (Rev 2) / `hermetic_webdav_run.md` (Rev 1) + research `FINDINGS.md`, all with synced HTML+PDF (0 fence-leaks §11.4.168). **Honest gate verdict (§11.4.6):** the clean zero-install promotion set is now COMPLETE (Cast-eureka+FTP+WebDAV); SFTP needs `sshd` (absent, no stdlib SSH server), SMB/NFS need samba/nfsd, `discovery_reflect.sh` scored leg hard-requires the absent `avahi-browse` client, ADB/container need device/podman — all genuinely operator-gated (§11.4.122/§11.4.3). **Email (SMTP-implicit-TLS + POP3S/IMAPS) is pure-stdlib-feasible and under active de-risking** — a standalone TLS-mail PoC subagent (`ac86a45fcd50747dd`) is in flight before any `hermetic_email_run.sh`. **Next: on PoC GO, build+review+land H2.email; the loop continues (no tag — "keep hardening"). Rev-13 wave (HEAD `b76065c`, branch `main`, all FF to origin): hermetic-harness H0-FULL DONE — a REAL encrypted kernel-WireGuard tunnel round-trip proven autonomously (`b76065c`).** The planned `wireguard-go` build was unnecessary: the host `wireguard` **kernel module** works inside an unprivileged userns netns (`ip link add` `wg0 type wireguard` + `wg set` succeed under `unshare -Ur`; `/usr/sbin/wg` present) — so H0-full is zero-build/zero-dep/no-package-install/no-podman/no-Mullvad/no-root. `tests/vpn_lan/hermetic_wg_roundtrip.sh`: veth underlay (10.9.0.x) carries the encrypted WG UDP, wg0 overlay (10.10.0.x) is the tunnel, a real HTTP payload served on the WG-only `10.10.0.2` and fetched over the tunnel — **`wg show`: latest-handshake=1782947082 rx=452 tx=752**, sha256 verified, **3/3 deterministic (§11.4.50)**; **golden-bad `WG_MUT=badkey` → hs=0/rx=0 → round-trip breaks** (§11.4.107(10)/§11.4.68 — `rx=0` proves the WG crypto gates the traffic, not the veth underlay). **Next: H1/H2** — run real peer services (smbd/vsftpd/webdav/eureka/mDNS, unprivileged) + wire `HELIX_BRIDGE_MODE=hermetic` into `tests/lib/svord_bridge.sh` so the protocol tests run INSIDE the namespace over the tunnel, promoting the 16 operator-gated SKIPs to AUTONOMOUS (§11.4.52). **Rev-12 wave (was HEAD `f96da56`, branch `main`, all FF to origin):** consolidation + a game-changing autonomous-validation path. `70f0636` **README §11.4.57 doc-link table completed** — all 6 Status pairs listed (the 2 VPN-LAN pairs were missing). `89cfd21` `.gitignore` `.tmp_export/` doc-export scratch (§11.4.30). **`1faead5` + `f96da56` — the hermetic-WireGuard test-harness path (deep research, §11.4.150/§11.4.52):** a loopback WireGuard pair in **unprivileged** network namespaces (`unshare -Ur -n` ⇒ root-in-userns + userspace `wireguard-go`, the `cmusatyalab/wireguard4netns` pattern) would let the **16 operator-gated protocol tests run AUTONOMOUSLY** against a controlled peer — **gated on NEITHER the broken podman NOR the live Mullvad bridge**. Design doc `1faead5` (FACT-probe: `kernel.unprivileged_userns_clone=1`, `max_user/net_namespaces=255793`, `unshare -Ur -n` rc=0, `/dev/net/tun` present). **H0 feasibility PROVEN with real physical evidence `f96da56`:** `tests/vpn_lan/hermetic_netns_poc.sh` — 2 netns + L3 veth + a real python3 HTTP payload served in the peer, fetched + **sha256-verified byte-for-byte, 3/3 deterministic (§11.4.50), golden-bad `POC_MUT=1` FAILs at the sha256 check (§11.4.107(10) teeth load-bearing)**, fully rootless, host-safe (§12, torn down with `unshare`). Honest scope (§11.4.6): veth not yet WireGuard — proves the substrate; real Mullvad topology stays a §11.4.3 operator-gated confirmation. Next: H0-full (build `wireguard-go`, swap veth→WG tunnel) — deferred pending host process-headroom (transient fork exhaustion observed §12). **Rev-11 wave (was HEAD `4e2810c`, all FF to origin):** the **VPN-LAN service-access feature is COMPLETE — all Phases 0-12 committed** incl. the operator's **Phase 12 bidirectional exposure** (`2ed0fed` bidirectional_exposure.md + ingress_allowlist_teeth.sh, wired into the standing suite) and **Phase 1 SSRF carve-out teeth** (suite-wired). **All 5 autonomous VPN-LAN teeth GREEN** (SSRF carve-out + SSRF_MUT, ingress-allowlist + INGRESS_MUT, bridge). Landed also: DNS-rebinding SSRF gap demo + smokescreen design (`4e2810c`, 7 sources §11.4.150), control-plane coverage **acl-helper 0→53.8%** (`4a9b75f`) + **cmd/api 0→61.1%**

(`4e2810c`, -race clean, go.sum untouched), §11.4.169 **stress+chaos** (`97d9733`, 100-iter identical hash) + **benchmark+memory** (`4e2810c`), deep **OSS survey** (`958d110`, 28 projects/50 URLs). **§11.4.1 anti-bluff win (`7c4345d+4a007a0+30ec5b5`):** the standing suite was HANGING forever on the LE phase3 guard (rootless-podman aardvark-dns bind failure); root-caused live (§11.4.102), fixed the LE boot-timeout→SKIP + three latent `set -e`/`pipefail`/`grep -c` suite-abort bugs → the suite now **COMPLETES + honestly reports (74/60/11/3)** instead of masking failures behind a hang. **HOST PODMAN BROKEN (env, operator-actionable, NOT a code defect):** `aardvark-dns` cannot bind netavark gateway `:53` host-wide → proxy containers show "Up (healthy)" but `crun` says NOT running, `:53128`/`:51080` don't serve, `./start` fails identically; a host-level podman/netavark reset is needed but was NOT done autonomously (§11.4.101/§11.4.174 — shared host w/ operator `lava-*/deploy_caddy/wg0-mullvad`). The 3 suite FAILs are ALL this one condition. **3 subagents in flight:** control-plane coverage r3, bidirectional both-way protocol assertions, §11.4.169 concurrency+load. **(Rev-10 prior:) §11.4.126 autonomous loop + VPN-LAN service-access feature (Phases 0/2/3/4/8/9 landed; 5/6/7/10/11 in flight)**. **Rev-10 wave (12 commits this session, all FF to origin, HEAD `d218977`):** `8140d40` S1 security-ACL SKIP→GREEN (authoritative access.log `TCP_DENIED/HIER_NONE`, standing suite 69/63/6/0, §11.4.169 matrix 10 PASS/1 SKIP); `fe9d9a9` VPN-LAN comprehensive phased plan (12 phases, env-var bridge §11.4.28, SSRF-reconciled §11.4.120, 5 cited deep-research streams §11.4.150); `12faf12` **Phase 8 Miracast §11.4.112 structurally-impossible verdict** (cited Wi-Fi-Alliance, Cast alternative); `628d255` `cmd/healthd` coverage 59.5→70.3% (`sample` 0→100%, -race clean, conductor-re-verified); `d781002` **Phase 0 env-var svord bridge scaffold** (`.env.example` + `tests/lib/svord_bridge.sh` + `scripts/svord_doctor.sh` + companion doc; 3 verdicts UP/SKIP/MISCONFIGURED reproduced); `a5e5616` **Phase 9 operator bridge-setup guide**; `5c28f56` **wired `test_vpn_lan_bridge` into the standing suite** (bridge-down honest SKIP + §11.4.115 teeth UP-stub⇒UP, suite GREEN **71/64/7/0**); `182e80a` **Phase 2/3 SMB/NFS + FTP/SFTP/WebDAV tests** (sha256 round-trip, wrong-answer⇒FAIL, bridge-down SKIP PASS_lines=0); `2f31460` **Phase 4 email + open-relay guard** (external-RCPT-accepted⇒FAIL, creds via stdin never argv); `d218977` **VPN-LAN integration Status + Status_Summary** (§11.4.45/§11.4.56). **Anti-bluff pattern proven:** every protocol test sources `tests/lib/svord_bridge.sh`, calls `bridge_require` FIRST, honest-SKIPs (exit 0, zero `^PASS:`) when the bridge is down — never a fake PASS; `ab_pass_with_evidence` refuses a PASS without a non-empty captured artefact. **3 subagents in flight (§11.4.70/§11.4.103):** Phase 6/7 Chromecast+ADB tests, Phase 11 Challenge + HelixQA `vpn_lan.yaml` bank, Phase 5 discovery-reflector design+test. **Data-plane health FACT (§11.4.7):** base proxy 204 on a real endpoint via `:53128`; the fake-domain `cache.example` 503/000 is NOT a regression. Host 43%. **(prior) §11.4.126 autonomous hardening loop** (operator: "keep hardening, don't tag yet"). **P10 VPN fail-closed = GREEN**: dynamic stack booted, tunnel DOWN → branded 503 `ERR_TUNNEL_DOWN` ×3, `leak_seen=0`, Squid PID unchanged, deterministic ×3 + RED-polarity; egress-half operator-gated SKIP (§11.4.21). **2 security fixes landed + VERIFIED DEPLOYED LIVE** (§11.4.108 runtime-signature): Squid header/version-hygiene (`via off`+`forwarded_for delete`+version-suppression — `Via` gone) `4f983ee`; Dante SOCKS5 SSRF (block link-local/loopback/RFC1918 + `command:connect`) `4626f05`. **Rev-7 hardening wave (5 commits):** `790c191` **S4 SSRF guard hardened** — the SOCKS-block verdict now requires dante's authoritative `block(N)` log line (§11.4.69), NOT elapsed-time (an independent §11.4.142 review found the timing-only discriminator bluff-capable on fast-refuse hosts); **security guard now WIRED into the standing suite** (`run-tests.sh test_security_guards`, §11.4.135, GREEN+RED-polarity, set-e-safe) — iterate-to-GO (§11.4.134); `487c918` **cache challenge authoritative** — reads Squid's own access.log via `podman exec` → real `TCP_MEM_HIT` (§11.4.69), no more SKIP-fallback (Squid caching proven genuine); `3755702` **control-plane unit coverage** store 64.5→98.2% / vpn 78.6→98.1% / api 69.6→80.3% / healthd 61.3→67.6% (real error/fail-closed branches, race-clean, go.sum unchanged); `86126ac` README §11.4.57 doc-link section; `ad720ec` this file → Rev 6. **2 items TRACKED-for-operator** (connectivity-risk §11.4.101): Squid `dns_nameservers` DNS-leak (static mode); Dante client-side open-relay. **§11.4.169 matrix = 9 PASS / 2 honest SKIP** (`docs/design/hardening/Status.md` Rev 3). **LE issuance + renewal BOTH PROVEN** — autonomous scope COMPLETE; Phase 4/6 OPERATOR-BLOCKED (§11.4.10). **Rev-8 increment (2 commits, subagent-driven §11.4.70 + conductor-reviewed pre-commit §11.4.142):** `017482a` closed the §11.4.18 companion-doc gap (16 of 61 scripts → `docs/scripts/<name>.md` + synced HTML+PDF, all 16 PDFs §11.4.168 leak-clean, library function-lists cross-checked against real source §11.4.6); `0e987f1` raised control-plane `internal/api` unit coverage 80.3→95.8% (real error/fail-closed/500/mTLS-bootstrap branches, race-clean, `go.mod`/`go.sum` unchanged). HEAD `0e987f1` (== `main` == github/origin/upstream). **Branch:** `main` (operator directive 2026-07-01: "all work merged to main, all future work on main") **Spec:** `docs/superpowers/specs/2026-06-30-vpn-aware-proxy-extension-design.md` (Rev 4) **Plan:** `docs/superpowers/plans/2026-06-30-vpn-aware-proxy-extension-plan.md` (Rev 1) **Authority:** Inherits the Helix Constitution submodule (`constitution/Constitution.md`) per §11.4.35.

> §12.10 live-state resume file. Read this first, then `git fetch --all --prune` and re-read `git log --oneline main..HEAD`. Any agent must be able to resume exactly where the last session left off from this single file.

---

1. Current PHASE

§11.4.126 autonomous hardening loop — construction (P0–P10 fail-closed) is landed; the loop is now closing hardening gaps under all §11.4.169 test types with real captured evidence **AND has opened a new feature workstream: VPN-LAN service access** (operator mandate — reach/use all mainstream services on the far side of the svord_toolkit VPN). The authoritative phased tracker is `docs/design/vpn_lan_access/PLAN.md` (§11.4.172, 12 phases). Phase 0 (env-var bridge scaffold + svord-doctor) is in flight via a subagent; Phase 8 (Miracast §11.4.112 verdict) + a control-plane coverage push run in parallel. Operator decision in force: **"keep hardening, don't tag yet."** The Go control-plane (stores, health-publisher, acl-helper, config-compiler, P5b breaker/failover, P6 control-API/SSE/metrics/PAC/mTLS) builds + vets + gofmt-clean and is proven unit / integration / config-parse; the `dynamic` compose profile boots live and the **fail-closed data-plane proof is GREEN**. Base proxy UP on `:53128` (204); host ~43%; 4 `helixproxy_*` Podman secrets present (enable dynamic re-boot).

P10 VPN fail-closed — GREEN (§11.4.68/.115/.108): `tests/dynamic/vpn_failclosed_test.sh` proven live — booted the dynamic stack, forced the tunnel DOWN via Redis `vpn:status`, tunnel-DOWN ⇒ branded 503 `ERR_TUNNEL_DOWN` ×3 (real 3132-byte page), `leak_seen=0`, Squid PID unchanged; deterministic ×3 (§11.4.50) + RED polarity guard FAILs a fabricated 200 leak (§11.4.115). Real-VPN-egress half is operator-gated SKIP (gluetun WireGuard creds §11.4.21). Evidence `qa-results/dynamic/vpn_failclosed/20260701T130115Z/`. Re-runnable boot recipe: 4 external Podman secrets from `tests/observability/gen_test_mtls.sh` → `./start --dynamic` (backgrounded) → poll `proxy-squid` Up + compiler renders `dynamic-routing.squid` → `HELIX_DYNAMIC_STACK=1 GOMAXPROCS=2 nice -n 19 ionice -c 3 bash tests/dynamic/vpn_failclosed_test.sh`; restore base after: `./stop && ./start`.

2 real security fixes (config-security review, `docs/design/security/Status.md` Rev 2) — RED→GREEN + sink-side evidence + standing guards:

- Squid header/version-hygiene (`4f983ee`): `via off` + `forwarded_for delete` + `httpd_suppress_version_string on` + `visible_hostname helix-proxy` in `squid.conf` + `squid.dynamic.conf`. The `Via: 1.1 proxy-squid (squid/6.13)` leak was CONFIRMED live (RED) → GONE (GREEN). Guard: `proxy_acl_security.sh` S3. **Root-cause note:** single-file `:ro` bind mounts pin the inode → config edits need a container **recreate** (`./stop && ./start`), NOT `squid -k reconfigure` (re-reads the stale inode).
- Dante SOCKS5 SSRF (`4626f05`): `command: connect` + `socks block` for 127/8, 169.254/16, 10/8, 172.16/12, 192.168/16 in `sockd.conf`. 5 internal targets refused fast (code 000 ~0.01s,

dante-log `block(N)`), external control 204, no public-egress regression. Guard: `proxy_acl_security.sh` S4.

2 items TRACKED-for-operator (§11.4.101 connectivity-risk, NOT autonomously fixed): (1) Squid `dns_nameservers 8.8.8.8` bypasses the DoT dnsproxy → DNS leak in **static** mode (dynamic mitigated by `never_direct`); re-point = risk. (2) Dante `socksmethod none` + `client pass from:0.0.0.0/0` open-relay if `:51080` escapes the bridge; client-CIDR restriction = risk. O(1) table in `docs/design/security/Status.md`.

§11.4.169 hardening matrix (`docs/design/hardening/Status.md` Rev 3) — 9 PASS / 2 honest SKIP: PASS = stress+chaos, DDoS(300/300), concurrency(40, crosstalk=0), memory(ratio 1.0017), **P10 fail-closed**, **race/deadlock (0 DATA RACE, 11 pkgs)**, **benchmark (200/200, p50=86ms/p95=88ms/p99=91ms, 10.84 req/s)** + unit/integration. SKIP (honest §11.4.3) = security ACL live-deny (no autonomous deny topology) + P10 egress-half (gluetun creds).

Remaining actionable (non-operator-gated) is thinning. Operator-gated: 2 TRACKED security items (connectivity risk), LE Phase 4/6, P10 real-egress (gluetun creds), HelixQA vendoring (6 un-vendored siblings), release tag.

2. Landed commits (newest first)

`main` FF-tracks HEAD (§11.4.113 FF-only), so `main..HEAD` is empty — HEAD == `main` == github/origin/upstream == `4dd5fc6`. The full feature-branch history since the original branch point is below; the historical table (numbered 1–28) is retained for the earlier construction wave.

Latest hardening wave (newest first):

Commit	Lane	Summary
`0e987f1`	control-plane	`internal/api` unit coverage 80.3→95.8% (real 404/400/500/502/fail-closed-TLS/metrics-suppression branches, race-clean, go.mod unchanged) — subagent-driven (§11.4.70), conductor-reviewed (§11.4.142)
`017482a`	docs	16 §11.4.18 companion docs (previously-undocumented test/challenge/lib scripts) + synced HTML+PDF, all §11.4.168 leak-clean, library fn-lists source-verified (§11.4.6)
`790c191`	security	S4 SSRF guard → authoritative dante `block(N)` log-line discriminator (§11.4.69) + security guard wired into standing

Commit	Lane	Summary
		suite (§11.4.135) — independent-review-driven (§11.4.142/.134)
`487c918`	challenge	proxy cache challenge → authoritative `TCP_*HIT` via container access.log (§11.4.69), no more SKIP-fallback (Squid caching proven genuine)
`3755702`	control-plane	unit coverage store 64.5→98.2% / vpn 78.6→98.1% / api 69.6→80.3% / healthd 61.3→67.6% (real error/fail-closed branches, race-clean)
`86126ac`	docs	README §11.4.57 Tracked-Items doc-link section (9 Status docs)
`ad720ec`	docs	CONTINUATION → Rev 6 (§12.10 sync: P10 GREEN + security fixes + §11.4.169 matrix)
`4626f05`	security	Dante SOCKS5 SSRF hardening — block internal/link-local/loopback egress + `command:connect` (RED→GREEN + S4 guard)
`4d0a7ed`	control-plane	drop dead nil-check in TestPostgresSatisfiesQueries (golangci SA4023)
`916e72b`	helixqa	unblock recipe for the proxy test bank (6 un-vendored own-org siblings)
`8833ccc`	letsencrypt	cert-analyzer edge-case coverage 37→55 (validity boundaries, malformed PEM, empty/IP/mixed SAN, double-wildcard)
`4f983ee`	security	Squid header/version-hygiene hardening (`via off` + `forwarded_for delete` + version suppression) — RED→GREEN + S3 guard
`1caaf51`	hardening	control-plane unit(100–61%)+Go-benchmarks+audit-atomicity-verified + Challenges 2/3 + §11.4.169 matrix sync
`567c9e1`	hardening	P10 VPN fail-closed GREEN + race(0)/ benchmark(p50=86ms) + §11.4.169 Status matrix

Earlier construction wave (historical, numbered from the original branch point):

#	Commit	Phase	Summary
28	`8d95f8a`	P6	real bidirectional metric-name drift guard (§1.1-mutation-proven) + concurrency consistency test; WARNING-3/4/5

#	Commit	Phase	Summary
27	`2bc03de`	BUGFIX-0006	revive + de-bluff `comprehensive-test.sh` (`(( ))` abort = 100% dead) + real B2/B3/B8 evidence; surfaced regression #50
26	`4394643`	BUGFIX-0005	`final-verify.sh` + `verify-proxy.sh` no longer green a NO-VPN config (false-VPN-routing §15) + `set -e` abort
25	`cd11494`	BUGFIX-0004	`run-tests.sh` no longer FAILs a healthy proxy — §11.4.3 topology-aware ports + 3-state SKIP
24	`6a8f886`	P11	refresh CONTINUATION to live state (Rev 2) — 23 commits, P5b/P6/BUGFIX-0002/0003 landed, P8 in flight
23	`61b4215`	chore	gofmt-format 6 pre-existing files (formatters-clean mandate; semantics-null verified)
22	`62b22fe`	P6	control-API server (REST/SSE/metrics/PAC, fail-closed mTLS) + coherent operator-wiring contract
21	`1045dfd`	BUGFIX-0003	`test_result` must `return 0` — suite no longer aborts mid-run under `set -e`
20	`c6f2935`	P9	§11.4.18 operator-guide companions for the 16 `tests/dynamic` scripts
19	`b5573a9`	BUGFIX-0002	squid log-dir writable under rootless Podman (proxy crash-loop) — existing features now serve live
18	`0aca034`	P9	anti-bluff dynamic-routing test/analyzer harness (`tests/dynamic`)
17	`e6e93ec`	P10-prep	`dynamic` compose profile + control-plane/squid Containerfiles + orchestrator wiring
16	`6bdeef9`	P5b	circuit-breaker + tier-failover (`internal/breaker`, gobreaker/v2)
15	`7d0d128`	P11	CONTINUATION (§12.10) + spec §9 reconcile + §11.4.65 HTML/PDF export backfill
14	`1833c8f`	P7.3	

#	Commit	Phase	Summary
			per-user Squid auth + rootless Podman-secret loader + kill-switch design (no secrets)
13	`603e039`	P5a	acl-helper — Squid external_acl OK/ERR from Redis, fail-closed (stdlib)
12	`e6e336f`	P7.1	per-tunnel DoH/DoT (dnsproxy) config plan + DNS-leak test design
11	`04526dd`	P4	config-compiler — render Squid/ Dante/PAC from PG + seed route keys (parse-verified)
10	`833fb9e`	P7.2	Prometheus scrape + Grafana dashboard config plan (promtool-validated)
9	`11106a4`	P3	vpn-health-publisher (cmd/ healthd + internal/vpn) — data-plane health, fail-closed, TDD
8	`b66d172`	P4	Squid 6.13 + Dante dynamic-mode templates (additive, parse-verified) + spec reconcile
7	`fbfe9ed`	P1	docs(spec): mark §20 gaps G1-G4 RESOLVED with spike decisions
6	`e19e0ed`	P2	store (pgx) + redis (go-redis) clients — fail-closed, TDD, real PG/Redis
5	`6409cb9`	P1	docs(research): resolve spec §20 gaps G1-G4 with captured-evidence spikes
4	`6802798`	P1/E	docs(audit): §11.4.138 forensic bluff-audit of 4 existing test scripts (8 bluffs)
3	`9ac1b4a`	P0	docs(dynamic-routing): DYNAMIC_ROUTING.md + 2 mermaid diagrams
2	`6251007`	P0	chore(submodules): incorporate containers, helix_qa, challenges, docs_chain (SSH, no-force)
1	`5f917a7`	P0	P0 scaffold — data model, evidence harness, Go skeleton, governance carriers

3. PROVEN-NOW vs OWED-TO-P10 (honest §11.4.6)

PROVEN-NOW (control-plane / config-plane / spike facts — captured)

- **Existing proxy serves LIVE (BUGFIX-0002)** — after the rootless-Podman log-dir fix, the booted `--no-vpn` stack proves all 3 existing features: HTTP forward proxy `200` + `Via: 1.1 proxy-squid`, Dante SOCKS5 `200`, squid cache `TCP_MEM_HIT` (no origin contact). Guard: `tests/regression/log_dir_writable_test.sh` (§11.4.115 polarity, §1.1 mutation byte-identical md5 `0128a96b6d467c2da5b7cef8a808e563`). Evidence: `qa-results/regression/bugfix38/`.
- **P5b breaker/failover** (`internal/breaker`, gobreaker/v2) — per-target circuit breaker + tunnel tier-failover, TDD.
- **P6 control-API** (`cmd/api` + `internal/api` + `internal/pac`) — REST CRUD + SSE + Prometheus `/metrics` + PAC, **fail-closed mTLS** (`RequireAndVerifyClientCert`), coherent operator-wiring contract (`CONTROL_API_TLS_CERT/_KEY/_TLS_CLIENT_CA`, `:58080`); builds + vets clean, §1.1 mutation md5 `67125c7a1ab9b00c98fb164f765b04af`.
- **Spec §20 gaps G1–G4 resolved** with transient-spike captured evidence (`docs/research/mvp/findings/F_spikes_G1-G4.md`, run-id `qa-results/spikes/20260630_205029_g1g4/`): G2 `ubuntu/squid:latest` = Squid **6.13** (not v8), §8 directive set `squid -k parse` exit 0; G4 gluetun **v3.40 (=v3.40.4)** control-API `:8000` answers 200, issue #3060 confirmed; G1 kernel-WG interface **creatable rootless** with `--cap-add NET_ADMIN`; G3 Dante **SIGHUP preserves an active SOCKS session** (20/20 chunks, curl exit 0, `/proc/net/tcp` ESTABLISHED proof).
- **P2 stores** (pgx + go-redis) — fail-closed, TDD, exercised against **real PG / Redis**.
- **P3 vpn-health-publisher** (`cmd/healthd` + `internal/vpn`) — data-plane health poll → Redis state, fail-closed, TDD.
- **P4 config-compiler + templates** — Squid 6.13 (`%>ha{Host}`) + Dante (concatenation, no `include`) render from PG; **`squid -k parse` exit 0**; PAC + route-key seeding parse-verified.
- **P5a acl-helper** — Squid `external_acl` OK/ERR from Redis, **fail-closed**, stdlib-only.
- **P7.2 observability** config plan — **promtool-validated** Prometheus scrape + Grafana dashboard.
- **P7.1 DNS / P7.3 security** — config plans only (DoH/DoT per-tunnel; per-user auth + Podman-secret loader + in-netns kill-switch). Design + parse layer.
- **Existing-test bluff audit** (Stream E) — 8 bluffs across 4 scripts catalogued (§11.4.138), guards owed to P8.

CAPTURED-AT-P10 (fail-closed data-plane proof — the dynamic stack HAS booted)

The `dynamic` compose profile (postgres + redis + control-plane + squid+helper + dante) now **boots live**; the fail-closed half of the usability proof is captured:

- `graceful_503` — **PROVEN**: tunnel DOWN (Redis `vpn:status`) → branded 503 `ERR_TUNNEL_DOWN` ×3 (3132-byte page) with **Squid PID unchanged**; deterministic ×3 + RED-polarity guard. Evidence `qa-results/dynamic/vpn_failclosed/20260701T130115Z/`.
- `no_leak` (tunnel-down case) — **PROVEN**: `leak_seen=0` during the DOWN window (no target reached while the tunnel is down).

OWED-TO-P10 (real-VPN-egress half — credential gate CLEARED 2026-07-02; proxy-data-plane e2e still owed)

UPDATE 2026-07-02 (§11.4.7): the gluetun-WireGuard-credential gate that blocked this whole section is **RESOLVED** — a persistent one-device Mullvad WireGuard identity is registered + stored (§11.4.10, gitignored `.env`), and **live Mullvad egress is PROVEN via gluetun** (rootless podman kernelspace WG tunnel, relay `cz-prg-wg-101`, `am.i.mullvad.net/json` → `mullvad_exit_ip=true`, exit IP `146.70.129.117` Prague CZ; evidence `qa-results/verification/mullvad_egress_20260702T161312Z/PROOF.txt`). What remains owed is the PROXY-data-plane half + the up→drop tcpdump killswitch:

- `vpn_real_egress` — **gluetun-level PROVEN** (`mullvad_exit_ip=true` above); still owed: egress IP **via the proxy data-plane** `== tunnel exit && != host IP` **+ `wg transfer` Δ** (200 OK is not routing).
- `no_leak / killswitch` (up→drop case) — drop a *real* tunnel → **zero** target packets on the real uplink (`tcpdump`) + DNS only via the intended resolver.
- per-user **407 auth challenge** live; **secret injection leak-free** at runtime.
- **G1 residual** — full rootless kernel-WG *operation* (handshake + routing + throughput), only interface *creation* was spiked (§20 G1).
- **G3 residual / P9** — concurrent / repeated SIGHUP + **route-change-mid-session** SOCKS path behaviour (§20 G3).
- circuit-breaker open → failover to next up tier — **landed** (`6bdeef9`, P5b); live-under-load failover proof still owed.

4. Remaining phases

Phase	Scope	State
P5b	per-target circuit breaker + tunnel tier-failover (`sony/gobreaker/v2`)	landed `6bdeef9`
P6	control-API + SSE + metrics + PAC + fail-closed mTLS	landed `62b22fe` (admin-UI templ/htmx + §11.4.170 host-rendered pixel proof = P6.2, deferred)
P8	fix existing-test bluffs → §11.4.3 topology dispatch / honest SKIP / §11.4.161 + §11.4.135 guards	landed (`cd11494/4394643/2bc03de/8d95f8a`)
P9	full test matrix + Challenges + HelixQA (all §11.4.169 types; G3 route-change-mid-session live test)	§11.4.169 matrix GREEN (9 PASS/2 SKIP); Challenges 2/3; HelixQA vendoring operator-gated (6 siblings)
P10	**live `dynamic`-mode boot + captured data-plane evidence = the usability proof**	**fail-closed half GREEN** (`567c9e1`); real-egress §11.4.21 credential gate **RESOLVED 2026-07-02** — persistent Mullvad WG identity stored (§11.4.10) + live gluetun egress PROVEN (`mullvad_exit_ip=true`, `qa-results/verification/mullvad_egress_20260702T161312Z/PROOF.txt`); proxy-data-plane-through-gluetun e2e = next step
P11	docs sync + HTML/PDF (+DOCX where mandated) exports (this CONTINUATION + .remember are part of it)	ongoing (this Rev 6 sync)
P12	whole-branch review (iterate-to-GO) + full retest + merge to `main` no-force +	operator-gated — "keep hardening, don't tag yet"

Phase	Scope	State
	prefixed release tag	

5. Binding constraints (non-negotiable)

- **Anti-bluff §11.4** — every PASS carries positive captured **data-plane** evidence; control-plane/config-parse green is necessary, never sufficient; the end-user-usability bar is met only at P10.
- **No force-push §11.4.113** — merge onto latest `main`, fast-forward only; force-push is forbidden with no exception.
- **Rootless Podman §11.4.161** — all containers rootless; no Docker-rootful, no sudo, no root escalation; orchestrate via the containers submodule (§11.4.76), build on the remote host (§11.4.173).
- **Secrets-as-names-only §11.4.10** — VPN creds / proxy-auth / mTLS keys via Podman secrets / file refs; **never** plaintext in git; `.env.example` documents refs only.
- **Operator-safe §11.4.174** — do **NOT** touch the operator's pre-existing resources: the host `wg0-mullvad` (UP kernel-WG) interface and any `lava-*` containers (e.g. `lava-postgres-thinker`) are off-limits; verify process/resource ownership before acting; block-don't-break on shared-host contention.
- **Host safety §12** — ≤60% memory (§12.6); no host power-state commands (CONST-033); pull images sequentially; `--rm` diagnostics; `df` first.

6. Resume now (next actionable)

1. `git fetch --all --prune` on **`main`** (operator: all work on main); confirm HEAD `860d38e` (== `main` == origin; integrate any newer foreign commit per §11.4.71, no force §11.4.113). The single canonical moment-valid resume file is `.remember/remember.md` (§11.4.131) — read it first. **Hermetic H0→H2 is DONE + HARDENED:** the H0-full real kernel-WireGuard tunnel (`hermetic_wg_roundtrip.sh`) + the model-A protocol promotions over it are all landed + independently reviewed + wired into the standing suite as §11.4.135 guards (`test_vpn_lan_hermetic` in `tests/run-tests.sh`) — **all four protocol legs + the H0 substrate are wired; email wired `b66b2fa` (task #66), closing the §11.4.6 gap the §11.4.169 ledger audit caught (independent review confirmed all 5 rows PASS in the real loop)**. All 5 harnesses carry the §11.4.111 wrong-destination negative control (proven load-bearing — bind-`0.0.0.0` ⇒ harness FAILs). **The clean zero-install promotion set is COMPLETE (§11.4.6): Cast-eureka + FTP + WebDAV + email** all run AUTONOMOUSLY over the tunnel (unmodified protocol tests, stdlib peers on `10.10.0.2`,

golden-bad teeth, 3/3 deterministic). Remaining protocol legs are genuinely operator-gated (§11.4.122/§11.4.3): SFTP (no `sshd`), SMB/NFS (no samba/nfsd), `discovery_reflect.sh` scored leg (no `avahi-browse` client), ADB (device), container (podman) — do NOT manufacture bluff harnesses for these. **The underlay-sniff AF_PACKET non-leak differential (task #63) is LANDED on the WG substrate** (`91af9c6`, `hermetic_wg_roundtrip.sh`, independent §11.4.142 review `a2b3c696` GO 11/11): during the positive round-trip it captures on the underlay `veth0` and asserts BOTH ciphertext present (type-4 `0x04` datagram to `:51820`) AND the per-run plaintext nonce ABSENT in raw underlay bytes; load-bearing golden-bad `SNIFF_MUT=plain` flips ONLY "plaintext absent" to FAIL (not a tautology, §11.4.107(10)), honest exit-3 on empty capture, honest `SNIFF-SKIP` when unavailable, 3/3 deterministic. FINDINGS §7/§7.1 → Rev 6 (IMPLEMENTED). **ALL ledger tasks #63-#76 LANDED/RESOLVED** (`91af9c6` underlay-sniff · `cdb0ccd` ethertype guard · `85d8b32` sniff fan-out · `b66b2fa` email-wiring · `172eb5c` orphan-test · `c32dcad` healthd coverage · `6e3e031+adca02e` **F1 fix + guard** · `5771a46` **#68** · **#67 COMPLETE** = DDoS `0c51f61` + soak `ceb4839` + chaos `159fcbc` · `3ec39d3` **#73** chaos no-leak vacuity · `c44e90c` **#74/#75** vpn_failclosed-reason + memory_soak-degenerate · `860d38e` **#76** assert_no_leak fail-closed — all independent §11.4.142 GO (subagent OR conductor-inline); **#69 Integration PASS** (`go test -run Integration -v` → 14 real podman Postgres/redis/gluetun PASS / 1 honest §11.4.3 SKIP / 0 FAIL, §11.4.6 correction of a prior PENDING) + **E2E honest §11.4.3 SKIP**, hardening Status Rev 6). **ANTI-BLUFF AUDIT LEDGER COMPLETE — 6 findings, all §11.4.118-discovered → RED-first fixed → §11.4.135-guarded:** F1 (#71/#72) · F-A/F-B (#73) · F-C/F-D (#74/#75) · F-E (#76); the core `tests/lib/evidence.sh` library is provably audited (8 verdict helpers — 7 fail-closed-solid + `assert_no_leak` F-E fixed). **The non-operator-gated actionable queue is now EXHAUSTED** — every tracked ledger task done, audit ledger closed, evidence library clean. **ALL remaining work is OPERATOR-GATED (surface per §11.4.66, never fake a PASS): live svord bridge** (biggest unlock — flips ~16 protocol SKIPs to live PASS + a full data-plane e2e), **LE #59** (Phase 4/6 credentials/DNS), **podman aardvark-dns compose-network** fix (restores the data-plane :53128/:51080 + the full-stack e2e), **real-device flash**, **HelixQA 6-sibling vendoring**. When re-engaging: a further §11.4.118 pass MAY audit the lower-traffic evidence.sh probes + the P10-pending analyzers (no live callers yet — no call-site bluff possible until wired). No release tag ("keep hardening"). Run ONE netns owner at a time (§11.4.119). Design `docs/design/vpn_lan_access/hermetic_wg_test_harness.md` Rev 2.

2. **Continue the §11.4.126 autonomous hardening loop** (operator: "keep hardening, don't tag yet"). Base proxy UP

`:53128` (204); ~43% host; 4 `helixproxy_*` Podman secrets present. Keep dispatching 3–4 parallel non-data-plane subagents on remaining actionable items (§11.4.103); the data plane / `:53128` has a single owner (§11.4.119) — coordinate before any boot.

3. **P10 fail-closed — GREEN + guarded (`567c9e1`).** Real-VPN-egress half remains operator-gated on gluetun WireGuard creds (§11.4.21/.66). To re-run fail-closed: 4 secrets from `tests/observability/gen_test_mtls.sh` → `./start --dynamic` (backgrounded) → `HELIX_DYNAMIC_STACK=1 … bash tests/dynamic/vpn_failclosed_test.sh` → restore base `./stop && ./start`.
4. **LE — issuance + renewal BOTH PROVEN (autonomous scope COMPLETE).** Phase 3 hermetic DNS-01 issuance + Phase 5 zero-downtime renewal/rotation are cert-analyzer-verified, re-runnable, and guarded (`tests/letsencrypt/phase3_issuance_guard.sh` + `phase5_rotation_guard.sh`, wired in `run-tests.sh`); custom Caddy image via `deploy/letsencrypt/build.sh`; cert-analyzer self-test 37→55 (`8833ccc`). Phase 4 (LE-staging token §11.4.10)
   - Phase 6 (prod domain) OPERATOR-BLOCKED, **reason "operator-deferred DNS (2026-07-02)"** — operator deferred live DNS 2026-07-02; unblock = operator-provided LE account + DNS credentials (§11.4.21 Operator-Block-Details in the Status doc). The hermetic Phase-3/5 guards stay GREEN. Docs `docs/design/letsencrypt/Status.md` (Rev 4).
5. **Operator-gated queue (surface, don't autonomously break):** 2 TRACKED security items (Squid `dns_nameservers` DNS-leak, Dante client-side open-relay — both connectivity-risk §11.4.101); **LE Phase 4/6 — reason "operator-deferred DNS (2026-07-02)"** (unblock = operator LE account + DNS creds); **P10 real-egress — §11.4.21 credential gate RESOLVED 2026-07-02** (Mullvad WG identity stored + live gluetun egress PROVEN; proxy-data-plane e2e still owed, §3 OWED-TO-P10); HelixQA vendoring (6 un-vendored siblings, `docs/helixqa/UNBLOCK.md`); the release tag `helix_proxy-0.1.0-dev-0.0.2` (operator said don't tag yet).
6. Every change: TDD reproduce-first (§11.4.43/§11.4.115), all warranted test types (§11.4.169), paired §1.1 mutation, independent review → iterate-to-GO (§11.4.142/§11.4.125/§11.4.134), docs in sync (§11.4.60/§11.4.65/§11.4.106), operator resources untouched (§11.4.174: `wg0-mullvad`, `lava-*`, `whoami:58080`).